



STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Roll Number	2520030607
Branch	CSE
Course Outcome	CO-1

Case Study Topic: MediFlow Hospital Patient Record Management using BST and AVL Tree

Work Done Summary:

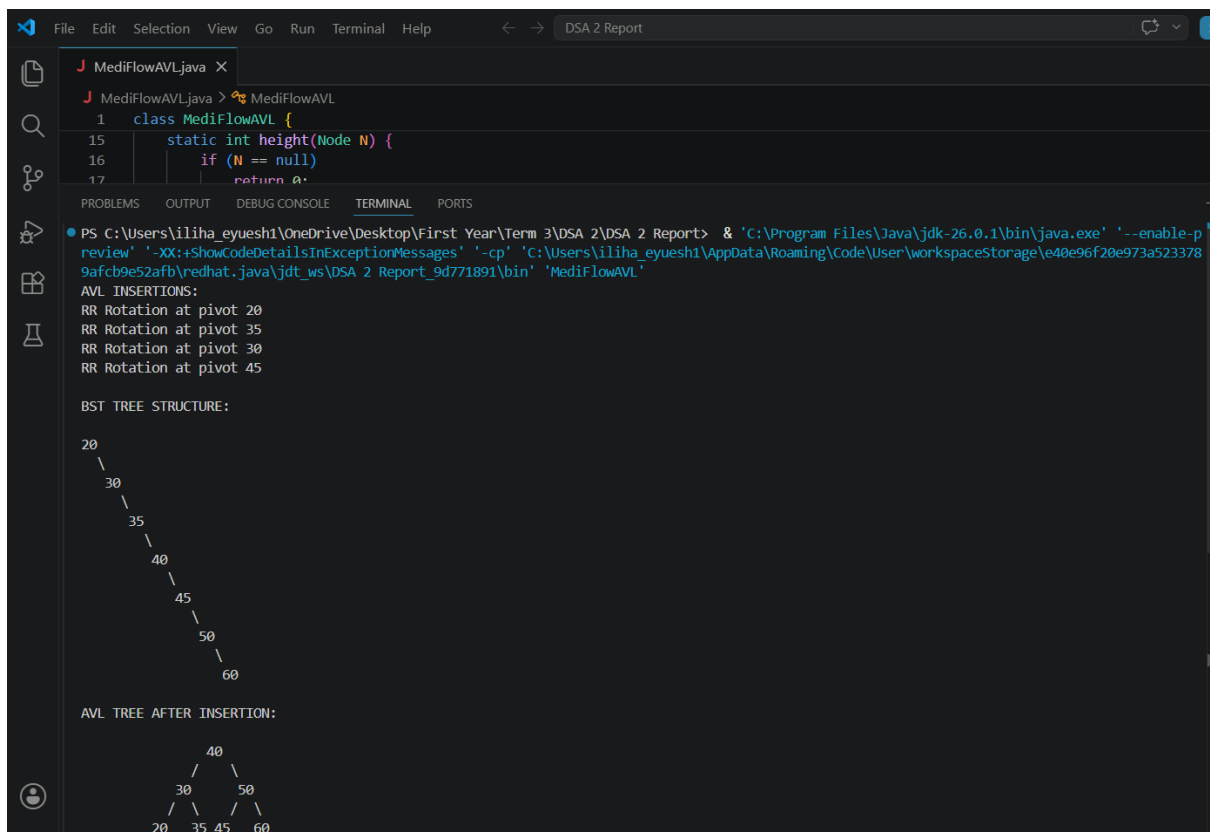
In this case study, we developed a patient record management system for MediFlow Hospital using Binary Search Tree (BST) and AVL Tree data structures. Patient records were organized using unique Patient IDs for quick access during emergency situations. Patient IDs 20, 30, 35, 40, 45, 50, and 60 were inserted in increasing order, which caused the BST to become unbalanced and inefficient. To improve performance, an AVL Tree was implemented, which automatically maintained balance through rotations. We constructed both BST and AVL Tree structures, identified pivot nodes during imbalance, deleted patient records with IDs 30 and 50, and compared the efficiency of BST and AVL Tree. The study showed that AVL Tree provides faster and more efficient search operations for hospital patient management.

Implementation Details:

1. Created a node structure to store Patient ID, left child, right child, and height.
2. Inserted patient IDs 20, 30, 35, 40, 45, 50, and 60 into a BST following standard insertion rules.

3. Observed that sorted insertion made the BST right-skewed and inefficient for searching.
4. Inserted the same IDs into an AVL Tree, where the balance factor was checked after every insertion.
5. When imbalance occurred, identified the pivot node and performed the required AVL rotations to restore balance.
6. Deleted patient IDs 30 and 50 from the AVL Tree and rebalanced the tree if needed.
7. Compared BST and AVL performance, concluding that AVL Tree provides faster and more efficient search operations.

Result/Output Screenshots:



The screenshot shows an IDE with a Java file named `MediFlowAVL.java` and a terminal window displaying the execution output.

Java Code:

```
1 class MediFlowAVL {  
15     static int height(Node N) {  
16         if (N == null)  
17             return 0;  
}
```

Terminal Output:

```
PS C:\Users\iliha_eyeshi\OneDrive\Desktop\First Year\Term 3\DSA 2\DSA 2 Report> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-p  
review' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\iliha_eyeshi\AppData\Roaming\Code\User\workspaceStorage\e40e96f20e973a523378  
9afcb9e52afb\redhat.java\jdt_ws\DSA 2 Report_9d771891\bin' 'MediFlowAVL'  
AVL INSERTIONS:  
RR Rotation at pivot 20  
RR Rotation at pivot 35  
RR Rotation at pivot 30  
RR Rotation at pivot 45  
  
BST TREE STRUCTURE:  
  
20  
 \ 30  
  \ 35  
   \ 40  
    \ 45  
     \ 50  
      \ 60  
  
AVL TREE AFTER INSERTION:  
  
      40  
     /  \  
    30   50  
   /  \ /  \  
  20 35 45 60
```

```

J MediFlowAVL.java x
J MediFlowAVL.java > MediFlowAVL
1 class MediFlowAVL {
15     static int height(Node N) {
16         if (N == null)
17             return 0;

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\iliha_eyuesh1\OneDrive\Desktop\First Year\Term 3\DSA 2\DSA 2 Report> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-p
AVL TREE AFTER DELETION:

      40
     /  \
    35   60
   /  \ /  \
  20  45

Final AVL Inorder Traversal:
20 35 40 45 60

AVL Tree maintained balance successfully.
PS C:\Users\iliha_eyuesh1\OneDrive\Desktop\First Year\Term 3\DSA 2\DSA 2 Report>

```

The BST became unbalanced because patient IDs were inserted in sorted order, making search operations slower. The AVL Tree automatically balanced itself using rotations whenever imbalance occurred. After deleting patient IDs 30 and 50, the AVL Tree remained balanced and continued to provide efficient searching.

Time Complexity:

- BST Search (worst case): $O(n)$
- AVL Insert/Delete/Search: $O(\log n)$

Compared to a normal BST, the AVL Tree performed better because it maintained balanced height, making it suitable for fast hospital patient record management.

GitHub Link : <https://github.com/iliharshika/DSA-2-Project>

Student Signature	Faculty Signature